Sample Efficient Search to Decision for kLIN

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Abstract

The kLIN problem concerns solving noisy systems of random sparse linear equations mod 2. It gives rise to natural candidate hard CSP distributions and is a cornerstone of local cryptography. Recently, it was used in advanced cryptographic constructions, under the name "sparse LPN".

For constant sparsity k and inverse polynomial noise rate, both search and decision versions of kLIN are statistically possible and conjectured to be computationally hard for $n \ll m \ll n^{k/2}$, where m is the number of k-sparse linear equations, and n is the number of variables.

We show an algorithm that given access to a distinguisher for $(k-1)$LIN with m samples, solves search kLIN with roughly $O(nm)$ samples. Previously, it was only known how to reduce from search kLIN with $O(m^3)$ samples, yielding meaningful guarantees for decision kLIN only when $m \ll n^{k/6}$.

The reduction succeeds even if the distinguisher has sub-constant advantage at a small additive cost in sample complexity. Our technique applies with some restrictions to Goldreich's function and kLIN with random coefficients over other finite fields.

Keywords: learning parity with noise, sparse LPN, local functions, random CSPs, search to decision, one-way functions, pseudorandom generators

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1 Introduction

The *learning parity with noise* problem asks to solve noisy random linear equations mod 2 [BFKL93]. The best known algorithms for it run in time $2^{O(n/\log n)}$, where n is the number of variables [BKW03]. The running time does not improve even if the task is relaxed to distinguishing the output of the noisy equations from random. Equivalence of search and decision is established by means of an efficient sample-preserving algorithm for finding a solution given access to an efficient distinguisher [AIK09].

A well-known variant requires equations to involve only a small number k of variables [Ale03]. Known as noisy kLIN, this "sparse learning parity with noise" problem lends itself to increased efficiency with no apparent loss in security, as long as the number m of equations satisfies $m \ll n^{k/2}$.

The advantages of sparse equations were first advocated by Goldreich, who put forward the study of one-way functions in which each output bit is a predicate of k input bits [Gol00]. Hardness of inversion depends on choice of the predicate and setting of parameters [MST06, BQ12, ABR16, AL18]. As in the dense case, one may ask whether the search task of inverting the candidate one-way function reduces to the decision task of distinguishing its output from random.

Applebaum reduced search of Goldreich's function with $m_s = O(m^3/\alpha^2)$ output bits to decision with m output bits and advantage α [App13]. The reduction applies to noisy predicates including kLIN. In the constant advantage regime, kLIN decision and search become easy as soon as m reaches $n^{k/2}$, so the guarantees are only meaningful when $m \ll n^{k/6}$. This has bearing on the sparsity, dimension, and number of output bits with which the function could be securely instantiated.

For the special case of kLIN, Bogdanov, Sabin, and Vasudevan [BSV19] obtained an efficient reduction with $m_s = O(m(m/\alpha)^{2/k})$, albeit one that only succeeds with $\exp(-\tilde{O}(m/\alpha)^{6/k})$ probability.

The noisy kLIN problem underlies the security has been the basis of a public-key encryption scheme by Applebaum, Barak and Wigderson [ABW10]. Interest in the problem has recently re-emerged following proposals for basing advanced cryptographic primitives on it [DIJL23, DJ24, CHKV24].

1.1 Our Results

We show a search to decision reduction for noisy kLIN: an algorithm that given access to a distinguisher for $(k-1)$LIN with m (sample, label) pairs, solves search kLIN with m_s (sample, label) pairs and success probability α_s, where m_s and α_s depend on the rate $\eta \in [0, 1)$ of the noise added to the labels and on the advantage $\alpha \in [0, 1]$ of the distinguisher. Specifically,

- the number of (sample, label) pairs for search is $m_s = O(nm + n/\eta^2\alpha^{2(k-2)})$,
- the success probability of search is $\alpha_s = 1 - \exp(-\Omega(\alpha^2 n)) + n\exp(-\Omega(\eta^2 m))$.

For constant k, α, and inverse polynomial noise rate η, this yields $m_s = O(mn)$. For $k = O(\log n)$, and constant α the number of samples grows polynomially, but even then gives meaningful search to decision tradeoffs. The reduction gives meaningful tradeoffs also for inverse polynomial α, depending on the magnitude of k. We highlight the following:

- Our reduction applies more generally to local functions with deterministic predicates of a certain natural form, covering well-studied instantiations of Goldreich's function.
- It is of interest even in the noiseless setting. Even though the search variant of noiseless kLIN can be solved by Gaussian elimination beyond the statistical threshold of $O(n)$ examples, the distinguishing algorithm when m exceeds $n^{k/2}$ is substantially simpler. It is an AC^0 algorithm, while Gaussian Elimination is believed to require more power than NC^1.
- It is a $\sqrt{n}$ factor away from being optimal in terms of the conjectured sample complexity for polynomial-time algorithms. In Section 6.1 we explain the difficulty in overcoming this barrier using our approach.

One could also consider what happens when the search algorithm has access to a *refuter* rather than to a distinguisher. A refuter is required to reject all "planted" kLIN instances and must accept a noticeable fraction of random ones. We show that in such a case, the sample complexity of the kLIN solver can be improved by an additional square factor in the advantage.

1.2 The kLIN Problem

Let $P\colon \{\pm 1\}^k \to \{\pm 1\}$ be a possibly randomized predicate. We are mainly interested in the noisy kLIN (also known as sparse LPN) predicate

$$P(z_1, \dots, z_k) = z_1 \cdots z_k \cdot e$$

with noise rate $\eta = \mathbb{E}[e] > 0$. Our results will also apply to random instances of Goldreich's function, in which case the predicate is deterministic.

An *instance* with secret $\mathbf{s} = (s_1, \dots, s_n) \in \{\pm 1\}^n$ is a collection of m independent examples consisting of a sample $\mathbf{i}$ and a label ℓ, each of the form

$$(\mathbf{i}; \ell) = (\mathbf{i}; P(\mathbf{s_i})) = (i_1, \dots, i_k; P(s_{i_1}, \dots, s_{i_k})), \tag{1}$$

where $\mathbf{i} = (i_1, \dots, i_k) \in [n]^k$ are independent indices sampled uniformly with replacement from $[n]$.

Our model deviates from the literature standard in that the indices $i_1, \dots, i_k$ are sampled with repetition. Some kLIN examples will therefore effectively contain fewer than k variables owing to cancellations. In Section 5 we explain why our main Theorem also extends in the more common model where each example depends on exactly k variables.

The *search* problem asks to find $\mathbf{s}$ from $(\mathbf{i}_1, \ell_1), \dots, (\mathbf{i}_m, \ell_m)$, and the *distinguishing* problem asks to tell apart the planted distribution (1) from the null distribution consisting of samples $\mathbf{i}_j$ with labels ℓ_j that are independent random bits.

There is no known efficient distinguisher of constant advantage when $m \ll n^{k/2}$, even though the distributions are statistically distinguishable provided $m > n/(1 - H(\eta))$. Search kLIN is also efficiently solvable with high probability when $m_s \geq K n^{k/2}$ for some constant K. One search algorithm with this sample complexity is described in Section 5.

The best known efficient distinguisher in the regime $m \ll n^{k/2}$ is somewhat model-dependent. In our model, the "collision finding" distinguisher that searches for repeated samples $(\mathbf{i}, \ell), (\mathbf{i}, \ell')$ and outputs $\ell \cdot \ell'$ has advantage $\Theta(\eta m^2 n^{-k})$ when $m < n^{k/2}$. This advantage is likely to be asymptotically optimal among all distinguishers of degree $\tilde{O}(n(n/m)^{2/(k-2)})$ for odd k [FKO06, GKM22].[1]

It is plausible that the sample complexity required to solve the search and decision variants is asymptotically the same. Our main result provides a partial explanation. If decision $(k-1)$LIN can be solved efficiently with advantage $\Omega(\max\{m^{-1/2(k-2)}, n^{-1/2}\})$ then search kLIN can be solved given $O(n)$ times more samples.

1.3 Theorem Statements

Our main result is the following search to distinguishing reduction for noisy kLIN.

Theorem 1. *There is an efficient algorithm that, given access to a distinguisher for $(k-1)$LIN for m examples with advantage α, solves search kLIN given $O(nm + n/\eta^2\alpha^{2(k-2)})$ examples except with probability $\exp(-\Omega(\alpha^2 n)) + n\exp(-\Omega(\eta^2 m))$, assuming the secret is uniform over $\{\pm 1\}^n$.*

The main new idea is described in Section 3, which shows a sample-efficient reduction from approximate search to distinguishing. Section 4 shows how to extract the actual secret from an approximate one. This transformation was proposed by Bogdanov and Qiao [BQ12] and used by Applebaum [App13] in the context of amplification. Here we optimize the sample complexity in the regime of subconstant advantage. The proof of Theorem 1 is in Section 5. Theorem 14 in Section 5 provides an extension to other finite fields in the high advantage regime.

Our result generalizes to any (possibly randomized) predicates of the form

$$P(z_1, \dots, z_k) = z_1 \cdot Q(z_2, \dots, z_k). \tag{2}$$

These include kLIN as a special case but also other studied examples [CEMT14, AL18].

Predicate Q admits a unique Fourier expansion [O'D14]

$$Q(z_2, \dots, z_k) = \sum_S \hat{Q}(S) \prod_{i \in S} z_i, \tag{3}$$

[1] A quirk of our model is that there is a better distinguisher when k is even: Look for the equation "0 = 1", i.e. an equation in which all the variables on the left cancel out. Such an equation is present with probability $\Theta(\eta m n^{-k/2})$. This distinguisher is ruled out in the distinct indices model.

with the Fourier coefficients given by $\hat{Q}(S) = \mathbb{E}\, Q(z_2, \dots, z_k) \prod_{i \in S} z_i$. The expectation is taken over both the (uniform) choice of $z_2, \dots, z_k$ and the potential randomness of Q.

Two parameters of interest are pure high degree $d(Q) = \min\{|S| : \hat{Q}(S) \neq 0\}$ and pure max-weight $\eta(Q) = \max_{S\colon |S|=d(Q)} |\hat{Q}(S)|$. Noisy $(k-1)$LIN has pure high degree $k-1$ and pure max-weight η.

Theorem 2. *There is an efficient algorithm that, given access to a distinguisher for predicate Q for m examples with advantage α, solves search for predicate P given $O(nm + n/\eta(Q)^2\alpha^{2(d(Q)-1)})$ examples except with probability $\exp(-\Omega(\alpha^2 n)) + n\exp(-\Omega(\eta(Q)^2 m))$, assuming the secret is uniform over all balanced strings of length n.*

The statement can be extended to almost balanced secrets, specifically secrets whose relative imbalance $|(s_1 + \cdots + s_n)/n|$ does not exceed $c_k \alpha^{d(Q)-1}$ for a sufficiently small constant c_k. Using arguments of Applebaum [App13] this is sufficient to obtain a variant of Theorem 2 for truly uniform secrets but with a messy dependence on α.

Theorem 1 can in fact be derived from Theorem 2. Our proofs of the two diverge in Section 4 because the proof of Theorem 1 is somewhat simpler and more direct so we present it first.

The distinguisher's advantage Theorems 1 and 2 are vacuous when the advantage of the distinguisher is $O(1/\sqrt{n})$. This is a natural limitation of our reduction, which makes only $O(n)$ black-box queries to the distinguisher.

In general, a q-query reduction to a $\alpha = o(1/\sqrt{q})$ advantage distinguisher cannot be expected to tell apart the planted from the null distribution (and therefore produce any distinguishing feature like a solution). The reason is that on any given query the distinguisher might reveal the correct answer with probability α and produce a random bit otherwise. The advantage of the reduction can then be no larger than the advantage in distinguishing an α-biased coin from a $-\alpha$-biased one from q flips, which is at most $O(q\alpha^2) = o(1)$.

The distinguisher's advantage can be amplified from α to α' at a multiplicative cost of $\Theta((\alpha'/\alpha)^2)$ in sample complexity [App13].

2 The Poisson sample model

It is more natural to state the search to distingushing reduction (Section 3) in the *Poisson sample model*. In this model the number of examples m is not a priori fixed. It is a Poisson random variable $\mathrm{Poisson}(\lambda n^k)$ of rate λn^k.

Equivalently, conditioned on the secret $\mathbf{s}$, the number of occurrences of each of the $2n^k$ possible sample-label pairs $(\mathbf{i}; \ell)$ is an independent Poisson random variable of rate $\lambda \cdot p_{\ell|\mathbf{s_i}}$. Here, $p_{\ell|\mathbf{z}}$ is the conditional probability that $P(\mathbf{z}) = \ell$ in the planted distribution and $p_{\ell|\mathbf{z}} = 1/2$ in the null distribution. We call λ the *rate per sample*.

In our parameter setting of interest, the rate per sample λ is on the order of m/n^k, which is much less than one. Most samples won't appear at all. Out of those that do most will appear once. The probability of a collision, i.e. of some sample appearing at least twice, is about $n^k\lambda^2/2 = \Theta(m^2/n^k)$.

The equivalence of the two descriptions is a consequence of the additivity of Poisson random variables: The sum of independent Poisson random variables of rates λ_τ is a Poisson random variable of rate $\sum \lambda_\tau$. Conversely, suppose each sample in a given collection is independently of type τ with probability p_τ. If the size of the collection is $\mathrm{Poisson}(\sum \lambda_\tau)$ then conditioned on the size, the samples are independently of type τ, each with probability proportional to λ_τ. In our application there are $2n^k$ types τ, one corresponding to each possible sample-label pair $(i_1, \dots, i_k; \ell)$.

The Poisson and fixed sample models are equivalent up to constant factors in (average) sample complexity and statistical error that is inverse exponential in the number of samples. Here are the conversion algorithms between the two models.

Fixed-to-Poisson The Poisson model with rate per sample $\lambda = m/2n^k$ is simulated by procedure *FtoP*: Choose the first M examples for a $\mathrm{Poisson}(m/2)$ random variable M provided that $M \leq m$, and fail if $M > m$.

Claim 3. *On input a fixed size m sample instance, FtoP produces an instance that is $e^{-m/8}$-statistically close to a Poisson instance of rate per sample $m/2n^k$.*

Poisson-to-fixed Conversely, the fixed number of samples model can be simulated by procedure *PtoF*: Choose a random subset of $m = \lambda n^k/2$ many (distinct) examples if at least this many are available and fail otherwise.

Claim 4. *On input a Poisson sample instance of rate per sample λ, PtoF produces an instance that is $e^{-\lambda n^k/12}$-close to a fixed sample instance of size $\lambda n^k/2$.*

Both claims follow from Poisson large deviation bounds [Can22, Theorem A.8]:

$$\Pr(\text{Poisson}(\lambda) \geq \lambda + x), \Pr(\text{Poisson}(\lambda) \leq \lambda - x) \leq \exp\bigl(-x^2/2(\lambda + x)\bigr). \tag{4}$$

Proof of Claim 3. If an infinite number of fixed samples was available and the "provided that $M \leq m$ condition was dropped the output would be perfectly Poisson of rate per sample λ. The statistical error is therefore bounded by the probability that the $M \leq m$ condition fails, namely that a Poisson$(m/2)$ random variable is greater than m. By (4) this probability is at most $\exp(-m/8)$. □

Proof of Claim 4. The number M of available examples is a Poisson(λn^k) random variable M. Conditioned on $M \geq m$ the output distribution perfectly follows that fixed number of samples model. The statistical error is therefore the probability that a Poisson(λn^k) random variable is less than $\lambda n^k/2$. By (4) this probability is at most $\exp(-\lambda n^k/12)$. □

3 Approximate search to distinguishing reduction

Our reduction uses the distinguisher to extract information about the equality of the secret entries s_1 and s_j for all $j \neq 1$. The reduction maps kLIN instances in which s_1 and s_j are equal and different into planted and random instances of $(k-1)$LIN, respectively. For this purpose the distinguisher is executed on the subset of examples that have s_1 or s_j as their first variables, suitably preprocessed. This basic reduction is implemented by algorithm S below. The advantage of working in the Poisson model is that the instances given from the distinguisher match one of the promised distributions (planted or random) perfectly.

The output of the reduction is an approximation $\hat{\mathbf{s}}$ of the secret $\mathbf{s}$. Our measure for the quality of approximation is the *overlap*

$$\langle \mathbf{s}, \hat{\mathbf{s}} \rangle = \frac{s_1\hat{s}_1 + \cdots + s_n\hat{s}_n}{n}.$$

Proposition 5. *If D is a distinguisher for Goldreich's function with predicate Q, rate per sample 2λ, and advantage α then the output of algorithm S below has average overlap at least α with the secret $\mathbf{s}$ given samples for predicate P with rate per sample λ, assuming the secret is uniformly random.*

Algorithm S: On input a multiset C of samples $(i_1, \ldots, i_k; \ell)$,

Guess the value $\hat{s}_1 \in \{\pm 1\}$ of s_1.
Set $A = \{(i_2, \ldots, i_k; \ell \cdot \hat{s}_1) \colon (1, i_2, \ldots, i_k; \ell) \in C\}$.
For $j = 2$ up to n,
  Set $B_j = \{(i_2, \ldots, i_k; \ell \cdot \hat{s}_1) \colon (j, i_2, \ldots, i_k; \ell) \in C\}$.
  Set $\hat{s}_j$ to equal the output of D on the collection of examples $A \cup B_j$.
Output $\hat{\mathbf{s}} = \hat{s}_1 \cdots \hat{s}_n$.

$A \cup B$ denotes union of multisets: If x occurs with multiplicities a and b in A and B, respectively, it occurs with multiplicity $a + b$ in $A \cup B$.

Given input $A \cup B_j$, the distinguisher has no information which examples came from A and which examples came from B_j. If (the guess is correct and) s_1 equals s_j, $A \cup B_j$ is identically distributed to a planted instance. If s_1 does not equal s_j the labels of the A-instances and B_j-instances take opposite values so $A \cup B_j$ is distributed identically to a random instance.

We view the distinguisher D as a function into the interval $[-1, 1]$. Its advantage is given by $(\mathbb{E}[D(\text{planted})] - \mathbb{E}[D(\text{null})])/2$.

Proof. We show that, assuming $\hat{s}_1 = s_1$, for every $j \geq 2$, $A \cup B_j$ is identical to a planted distribution sample when $s_j = s_1$ and to a null distribution sample when $s_j \neq s_1$. By our assumption on D and the uniformity of s_j (given s_1),

$$\mathbb{E}[\hat{s}_j s_j] = \frac{1}{2}\,\mathbb{E}[\hat{s}_j s_1 | s_j = s_1] - \frac{1}{2}\,\mathbb{E}[\hat{s}_j s_1 | s_j \neq s_1] = \frac{1}{2}\,\mathbb{E}[\text{planted}] - \frac{1}{2}\,\mathbb{E}[\text{null}] = \alpha.$$

Therefore $\hat{\mathbf{s}}$ and $\mathbf{s}$ have average overlap $(1 + (n-1)\alpha)/n \geq \alpha$.

Assuming $\hat{s}_1 = s_1$, collection A is a Poisson sample for Goldreich's function with predicate Q with planted assignment s rate per sample λ. If s_j and s_1 are equal so is B_j. By additivity, $A \cup B_j$ is a Poisson sample for the planted distribution with rate per sample 2λ.

If s_2 and s_1 are not equal then each sample-label pair $(i_2, \ldots, i_k; \ell) \in [n]^{k-1} \times \{\pm 1\}$ has Poisson(λp) occurrences in collection A and Poisson$(\lambda(1-p))$ occurrences in collection B_j, where $p = p_{\ell | s_{i_2}, \ldots, s_{i_k}}$. By additivity, $(i_2, \ldots, i_k; \ell)$ occurs Poisson$(\lambda p + \lambda(1-p))$ = Poisson(λ) times, independently among all such pairs. This is identical to the null distribution with rate per sample 2λ. □

Proposition 5 guarantees *average* overlap α between the output of S and the secret. If α was as large as $1 - \delta/n$ this would be sufficient to recover the secret except with probability δ. Such large advantage can be attained by amplification at a *multiplicative* cost in sample complexity. We pursue an alternative route that entails only an additive cost, provided α is at least $1/\sqrt{n}$.

When α is small, it could be that this advantage is lopsided in the sense that in a $(1-\alpha)$ fraction of instances S has no advantage at all and it completely recovers the secret in the rest.

The following variant of the reduction ensures that S has good advantage on a large fraction of instances provided α is not too small. A secondary difference is that it also applies to secrets that are balanced, which is the setting of our Theorem 2.

Theorem 6. *If D is a secret-independent distinguisher for Goldreich's function with predicate Q, rate per sample 2λ, and advantage α then for at least one $i^\star \in [n]$ the output of algorithm $S^\star$ below has overlap at least $\Omega(\alpha)$ with the secret for all but a $3\exp(-\Omega(\alpha^2 n))$ fraction of instances, given samples for predicate P with rate per sample λ, assuming the secret is uniformly random or a uniformly random balanced string.*

A distinguisher is *secret-independent* if its advantage does not depend on the secret conditioned on the samples. Namely, for every pair of secrets $\mathbf{s}$ and $\mathbf{s}'$,

$$\mathbb{E}[D(\mathbf{i}_1, Q(\mathbf{s}_{\mathbf{i}_1}), \ldots, \mathbf{i}_m, Q(\mathbf{s}_{\mathbf{i}_m}))] = \mathbb{E}[D(\mathbf{i}_1, Q(\mathbf{s}'_{\mathbf{i}_1}), \ldots, \mathbf{i}_m, Q(\mathbf{s}'_{\mathbf{i}_m}))].$$

In the proofs of the main theorems secret-independence is enforced by rerandomization.

The algorithm $S^\star$ is the same as S with variable $s_{i^\star}$ playing the role of s_1.

Algorithm $S^\star$: On input a multiset C of samples $(\mathbf{i}, \ell)$,

1 Guess the value $\hat{s}_{i^\star} \in \{\pm 1\}$ of $s_{i^\star}$.
2 Set $A_i = \{(i_2, \ldots, i_k; \ell \cdot \hat{s}_{i^\star}) \colon (i, i_2, \ldots, i_k; \ell) \in C\}$.
3 For $j \neq i^\star$, set $\hat{s}_j = D(A_{i^\star} \cup A_j)$.
4 Output $\hat{\mathbf{s}} = \hat{s}_1 \cdots \hat{s}_n$.

Proof. For an (unlabeled) sample collection $C = \{\mathbf{i}_1, \ldots, \mathbf{i}_m\}$ let $adv(C)$ be the advantage of the distinguisher on C:

$$adv(C) = \mathbb{E}[D(\mathbf{i}_1, Q(\mathbf{s}_{\mathbf{i}_1}), \ldots, \mathbf{i}_m, Q(\mathbf{s}_{\mathbf{i}_m}))] - \mathbb{E}[D(\text{null})].$$

By the secret-independence assumption, $adv(C)$ does not depend on $\mathbf{s}$.

If A and B are independent collections of examples (sample-label pairs) with sample rate λ respectively, by assumption $adv(A \cup B)$ equals α. By Markov's inequality, for at least a $\alpha/2$ fraction of multisets A, the advantage must be at least $\alpha/2$ over the choice of B (conditioned on A). Call such a multiset A good.

Assume for now that $\mathbf{s}$ is uniformly random. Let I and J be an arbitrary partition of the variables with $|I| = \alpha n/10$. As each A_i is good with probability at least $\alpha/2$ and these events are independent across $i \in I$, there exists some $i^\star \in I$ for which $A_{i^\star}$ is good except with probability at most $(1-\alpha/2)^{|I|} \leq \exp(-\alpha|I|/2) = \exp(-\alpha^2 n/20)$.

Assuming $A_{i^\star}$ is good, by the argument in the proof of Proposition 5, for every $j \in J$, $\hat{s}_j = s_j$ with conditional probability at least $\alpha/2$. These events are now mutually independent conditioned on $A_{i^\star}$. By Hoeffding's bound, $\sum_{j\in J} s_j \hat{s}_j$ is at least $\alpha|J|/4$ except with probability $\exp(-\Omega(\alpha^2|J|)) = \exp(-\Omega(\alpha^2 n))$. If this happens the total overlap $\langle \hat{\mathbf{s}}, \mathbf{s}\rangle$ is at least

$$\frac{1}{n}\cdot\left(\frac{\alpha|J|}{4} - |I|\right) = \frac{1}{n}\cdot\left(\frac{\alpha n}{4} - (1+\alpha/4)|I|\right) \geq \frac{1}{n}\cdot\left(\frac{\alpha n}{4} - \frac{5}{4}|I|\right) = \frac{1}{n}\cdot\left(\frac{\alpha n}{4} - \frac{\alpha n}{8}\right) = \frac{\alpha}{8}.$$

If $\mathbf{s}$ is uniform over balanced strings the partition of variables into I and J is no longer arbitrary. Let P be a Binomial($(1-\alpha/10)n, 1/2$) random variable. If both P and $1-P$ are at most $n/2$ then J is set to contain some P variables of value $+1$ and some $(1-\alpha/10)n - P$ variables of value -1. The remaining variables are put in I. (If not the variables are partitioned arbitrarily.) The assignment to $x_j : j \in J$ is a uniformly random string except with the probability that $\max\{P, (1-P)\} > n/2$. By a Chernoff bound this is at most $2\exp(-\Omega(\alpha^2 n))$ and is subsumed into the failure probability. □

4 Amplification of overlap

We show how to recover the secret $\mathbf{s}$ in a planted instance given an α-overlapping approximation $\hat{\mathbf{s}}$. Specializing to kLIN, the idea is to predict secret entry s_{i_1} by the parity $s_{i_2}\cdots s_{i_k}\ell$ and amplify the prediction advantage to almost 1 using sufficiently many independent examples. By Proposition 10 below $\Omega(\alpha^{2(k-1)}\log n)$ examples suffice for this.

In Theorem 11 we improve the dependence of sample size on α. To accomplish this the advantage is doubled in stages until it becomes constant at which point Proposition 10 is applied with constant α. In Theorem 13 we extend the argument to other predicates of type (2), provided the secret is balanced.

4.1 Noisy parities

We specialize Q to the noisy parity predicate $z_2\cdots z_{k-1}e$ with $\mathbb{E}\, e = \eta$. We use $\mathbf{s}$ to denote the secret of k-LIN instance C throughout this section.

Algorithm *Amp***:** On input a multiset C of samples $(\mathbf{i}, \ell)$, and $\hat{\mathbf{s}} \in \{\pm 1\}^n$

For every $i \in [n]$,

  Let s'_i be the majority of $\hat{s}_{i_2}\cdots\hat{s}_{i_k}\cdot\ell$ over $(i, i_2, \ldots, i_k; \ell) \in C$.

Output $\mathbf{s}' = s'_1 \cdots s'_n$.

Proposition 7. *There exists a constant $c > 0$ so that assuming $0 \leq \langle \mathbf{s}, \hat{\mathbf{s}}\rangle = \alpha < c/3$, the output $\mathbf{s}'$ of Amp satisfies $\langle \mathbf{s}, \mathbf{s}'\rangle \geq 2\alpha$ except with probability $\exp(-\Omega(\alpha^2 n))$, provided C contains $9/c^2\eta^2\alpha^{2(k-2)}$ samples whose first entry is i for every $i \in [n]$.*

Claim 8. *Given a k-noisy parity example $(\mathbf{i}, \ell)$ with secret $\mathbf{s}$, and an $\hat{\mathbf{s}}$ such that $\langle \mathbf{s}, \hat{\mathbf{s}}\rangle = \alpha \geq 0$, $\mathbb{E}[s_{i_1}\cdot \ell\hat{s}_{i_2}\cdots\hat{s}_{i_k} \mid \mathbf{s}, \mathbf{s}'] = \eta\cdot\alpha^{k-1}$.*

Proof. The label ℓ equals $s_{i_1}s_{i_2}\cdots s_{i_k}e$. As the pairs $(s_i, \hat{s}_i)$ are mutually independent across i, s_{i_1} cancels out and the expectation equals $\mathbb{E}[s_{i_2}\hat{s}_{i_2}|\mathbf{s},\mathbf{s}']\cdots\mathbb{E}[s_{i_k}\hat{s}_{i_k}|\mathbf{s},\mathbf{s}']\,\mathbb{E}[e] = \eta\alpha^{k-1}$. □

Claim 9. *There exists a constant $c > 0$ such that for every $\alpha > 0$ and odd n, $\mathbb{E}\,\mathrm{sign}(X_1 + \cdots + X_n) \geq \min\{c\alpha\sqrt{n}, c\}$, where $X_1, \ldots, X_n \in \{\pm 1\}^n$ are independent and α-biased.*

Proof. As the expectation increases monotonically in α, it is sufficient to prove the claim under the assumption $\alpha\sqrt{n} \leq 1$. Theorem 5.19 in [O'D14] is the identity

$$\mathbb{E}\,\mathrm{sign}(X_1 + \cdots + X_n) = \frac{2n}{2^n}\binom{n-1}{(n-1)/2}\sum_{k\text{ odd}}(-1)^{(k-1)/2}\cdot\frac{1}{k}\cdot\binom{(n-1)/2}{(k-1)/2}\cdot\alpha^k.$$

The leading term $k = 1$ dominates the sum. The total contribution of the other terms can be generously upper bounded by

$$\sum_{k\,\geq\,3\text{ odd}}\binom{(n-1)/2}{(k-1)/2}\cdot\alpha^k = \alpha\big((1+\alpha^2)^{(n-1)/2} - 1\big) \leq \alpha\left(\left(1+\frac{1}{n}\right)^{n/2} - 1\right) \leq 0.64\alpha.$$

By Stirling's approximation formula,

$$\mathbb{E}\,\mathrm{sign}(X_1 + \cdots + X_n) \geq 0.36\alpha \cdot \frac{2n}{2^n}\binom{n-1}{(n-1)/2} = \Omega(\alpha\sqrt{n}). \qquad \square$$

Proof of Proposition 7. Let c be the constant from Claims 8 and 9. By these two claims, for every i, either $\mathbb{E}[s_i s_i'] \geq c \geq 3\alpha$, or

$$\mathbb{E}[s_i s_i'] \geq c \cdot \eta\alpha^{k-1} \cdot \sqrt{\frac{9}{c^2\eta^2\alpha^{2(k-2)}}} = 3\alpha.$$

The statistics $\mathbb{E}[s_i s_i']$ are independent and at least 3α-biased across i. By Hoeffding's bound, $\langle \mathbf{s}, \mathbf{s}' \rangle \geq 2\alpha$ except with probability $\exp(-\Omega(\alpha^2 n))$. $\square$

Proposition 10. *If $\langle \mathbf{s}, \hat{\mathbf{s}} \rangle \geq c$, the output of Amp equals $\mathbf{s}$ except with probability $n\exp(-\Omega(\eta^2 c^{2(k-1)} r))$, provided C contains at least r samples whose first entry is i for every $i \in [n]$.*

Proof. By Claim 8 and the Hoeffding bound, $\Pr[s_i \neq \hat{s}_i]$ is at most $\exp(-\Omega(\eta^2 c^{2(k-1)} r))$ for every i. The proposition follows by a union bound over i. $\square$

Theorem 11. *Assuming $\langle \mathbf{s}, \hat{\mathbf{s}} \rangle \geq \alpha$, algorithm Recover outputs $\mathbf{s}$ except with probability $\exp(-\Omega(\alpha^2 n)) + n\exp(-\Omega(\eta^2 r))$, provided C contains at least $K/\eta^2\alpha^{2(k-2)} + r$ samples whose first entry is i for every $i \in [n]$ for some constant K.*

We say a set of samples is α-sufficient if it contains $9/c^2\eta^2\alpha^{2(k-2)}$ samples whose first entry is i for every i.

Algorithm *Recover***:** On input a multiset C of samples $(\mathbf{i}, \ell)$, and $\hat{\mathbf{s}} \in \{\pm 1\}^n$, and $\alpha > 0$

While $\alpha < c/3$,
  Replace $\hat{\mathbf{s}}$ by the output of *Amp* on a fresh α-sufficient subsample of C.
  Double α.
Output *Amp* on a fresh r subsample of C.

Proof. By Proposition 7, the invariant $\langle \mathbf{s}, \hat{\mathbf{s}} \rangle \geq \alpha$ is maintained throughout the while loop except with probability $\exp(-\Omega(\alpha^2 n)) + \exp(-\Omega((2\alpha)^2 n)) + \cdots = \exp(-\Omega(\alpha^2 n))$. By the end of the loop the overlap is then at least $2c/3$. Since we can make c arbitrary close to 1, by Proposition 10 the output then equals $\mathbf{s}$ except with additional probability $n\exp(-\Omega(\eta^2 r))$.

The required number of samples that start with i is

$$\frac{9}{c^2\eta^2} \cdot \left(\frac{1}{\alpha^{2(k-2)}} + \frac{1}{(2\alpha)^{2(k-2)}} + \cdots\right) + r \leq \frac{18}{c^2\eta^2\alpha^{2(k-2)}} + r. \qquad \square$$

4.2 Goldreich's function

We generalize Theorem 11 to arbitrary Q of type (2). The analogue of Proposition 7 is

Proposition 12. *There exists a constant $c > 0$ so that assuming $\mathbf{s}$ is balanced and $0 \leq \langle \mathbf{s}, \hat{\mathbf{s}} \rangle = \alpha < c/3$, the output $\mathbf{s}'$ of Amp satisfies $\langle \mathbf{s}, \mathbf{s}' \rangle \geq 2\alpha$ except with probability $\exp(-\Omega(\alpha^2 n))$, provided C contains $9/c^2\eta(Q)^2\alpha^{2d(Q)}$ samples whose first entry is i for every $i \in [n]$.*

Let $S^\star$ be the set of size $d(Q)$ (the pure high degree of Q) that maximizes $|\hat{Q}(S)|$ and $\sigma = \mathrm{sign}\,\hat{Q}(J)$.

Algorithm *Amp***:** On input a multiset C of samples $(\mathbf{i}, \ell)$, and $\hat{\mathbf{s}} \in \{\pm 1\}^n$

For every $i_1 \in [n]$,
  Let s_{i_1}' be the majority of $\ell \cdot \sigma \cdot \prod_{j \in S^\star} \hat{s}_{i_j}$ over $(i_1, i_2, \ldots, i_k; \ell) \in C$.
Output $\mathbf{s}' = s_1' \cdots s_n'$.

Proof. We show the analogue of Claim 8: $\mathbb{E}\, s_{i_1} \ell\sigma \prod_{j\in S^\star} \hat{s}_{i_j} \geq \eta\alpha^d$. The conclusion then follows from Claim 9 as in the proof of Proposition 7.

Applying the Fourier expansion (3) of Q,

$$\begin{aligned}\mathbb{E}\, s_{i_1} \cdot \ell \cdot \sigma \cdot \prod_{j\in S^\star} \hat{s}_{i_j} &= \sum_S \sigma \hat{Q}(S)\, \mathbb{E} \prod_{j\in S} s_{i_j} \prod_{j\in S^\star} \hat{s}_{i_j} \\ &= \sigma \hat{Q}(S^\star) \prod_{j\in S^\star} \mathbb{E}\, s_{i_j} \hat{s}_{i_j} - \sigma \sum_{S\neq S^\star} \hat{Q}(S) \cdot \mathbb{E} \prod_{j\in S} s_{i_j} \prod_{j\in S^\star} \hat{s}_{i_j}.\end{aligned}$$

As σ is the sign of $\hat{Q}(S^\star)$ the leading term equals $\eta\alpha^{|S^\star|} = \eta(Q)\alpha^{d(Q)}$. All other terms vanish: By independence,

$$\mathbb{E} \prod_{j\in S} s_{i_j} \prod_{j\in S^\star} \hat{s}_{i_j} = \prod_{j\in S\setminus S^\star} \mathbb{E}\, s_{i_j} \cdot \prod_{j\in S\triangle S^\star} \mathbb{E}\, s_{i_j} \hat{s}_{i_j} \cdot \prod_{j\in S^\star\setminus S} \mathbb{E}\, \hat{s}_{i_j}.$$

As $S \neq S^\star$ and $|S| \geq |S^\star|$ by the minimality of $S^\star$, the first product term is nonempty. Since $\mathbf{s}$ is balanced it evaluates to zero. □

The analogue of Theorem 11 is

Theorem 13. *Assuming* $\mathbf{s}$ *is balanced and* $\langle \mathbf{s}, \hat{\mathbf{s}} \rangle \geq \alpha$, *algorithm Recover outputs* $\mathbf{s}$ *except with probability* $\exp(-\Omega(\alpha^2 n)) + n\exp(-\Omega(\eta(Q)^2 r))$, *provided* C *contains at least* $K/\eta(Q)^2\alpha^{2(d(Q)-1)} + r$ *samples whose first entry is* i *for every* $i \in [n]$ *for some constant* K.

5 Proofs of Main Theorems

We compose the approximate search to distinguishing reduction and the amplification to prove Theorems 1 and 2. We also explain why Theorem 1 extends to the more common random kLIN model in which the sample indices $i_1, \ldots, i_k$ are distinct.

Proof of Theorem 1. Partition the instance into two parts, C_1 of size Poisson(mn) and C_2 in which for every $i \in [n]$ there are at least $K/\eta^2\alpha^{2(k-2)} + m$ samples that start with i. By Claim 3 and the Hoeffding bound the samples in C suffice for C_1 and C_2 except with probability $\exp(-\Omega(m))$.

To recover the secret, $S^\star$ is executed on C_1 to produce $\hat{\mathbf{s}}$. Then *Recover* is executed on C_2 and $\hat{\mathbf{s}}$ to recover the secret.

By Theorem 6, $\hat{\mathbf{s}}$ has $\Omega(\alpha)$ overlap with the secret except with probability $\exp(-\Omega(\alpha^2 n))$, provided it is given access to a secret-independent $(k-1)$LIN distinguisher for Poisson$(2m)$ many samples. Secret-independence is enforced by rerandomizing the secret, namely by shifting every label by $r_{i_1} \cdots r_{i_k}$ for a random secret r. This transformation preserves distinguishing advantage (under a uniform choice of secret). By Claim 4 a distinguisher for Poisson$(2m)$ samples can be realized from a distinguisher for m samples except with probability $\exp(-\Omega(m))$. As the distinguisher is invoked by $S^\star$ less than n times the overlap is as promised except with probability $n\exp(-\Omega(m))$.

Applying *Recover* on inputs C_2 and $\hat{\mathbf{s}}$ (which is independent of C_2) with $r = m$, by Theorem 11 the secret is recovered except with probability $\exp(-\Omega(\alpha^2 n)) + n\exp(-\Omega(\eta^2 m))$. □

The main difference in the proof of Theorem 2 is in the enforcement of secret-independence. The transformation now consists of a random permutation of the sample entries, namely example $(i_2, \ldots, i_k; \ell)$ is replaced by $(\pi(i_2), \ldots, \pi(i_k); \ell)$ for a random permutation π on $[n]$. As

$$(\pi(i_2), \ldots, \pi(i_k); Q(s_{i_2}, \ldots, s_{i_k}))$$

is identically distributed to

$$(i_2, \ldots, i_k; Q(s_{\pi^{-1}(i_2)}, \ldots, s_{\pi^{-1}(i_k)})),$$

the Hamming weight of $\mathbf{s}$ is a sufficient statistic of the permuted sample, so the distinguisher's advantage does not depend on $\mathbf{s}$ among balanced strings. The other difference is that Theorem 13 is used in lieu of Theorem 11 to analyze the amplification.

5.1 Extension to exact weight instances

In the exact weight kLIN model the sample elements $(i_1, \dots, i_k)$ are chosen without repetition. One complication in the search to distinguishing reduction, for instance algorithm S, is that the samples of $A \cup B_j$ are no longer identically distributed as those of a random $(k-1)$LIN instance. Owing to the lack of repetitions, the average number of occurrences of indices 1 and j ("variables" s_i and s_j) is about half that of the other variables. The distinguisher is not guaranteed to provide any useful information on such an atypical instance. To mitigate this effect the reduction discards those examples in A that include index j and those examples in B_j that include index 1.

For simplicity we describe the modification of algorithm S to the exact weight setting. The modification to $S^\star$ is analogous.

Algorithm S**:** On input a multiset C of samples $(i_1, \dots, i_k; \ell)$,

1 Guess the value $\hat{s}_1 \in \{\pm 1\}$ of s_1.

2 For $j = 2$ up to n,

3 Set $A_j = \{(i_2, \dots, i_k; \ell \cdot \hat{s}_1) \colon (1, i_2, \dots, i_k; \ell) \in C, j \notin \{i_2, \dots, i_k\}\}$.

4 Set $B_j = \{(i_2, \dots, i_k; \ell \cdot \hat{s}_1) \colon (j, i_2, \dots, i_k; \ell) \in C, 1 \notin \{i_2, \dots, i_k\}\}$.

5 Set $\hat{s}_j$ to equal the output of D on the collection of examples $A_j \cup B_j$.

6 Output $\hat{\mathbf{s}} = \hat{s}_1 \cdots \hat{s}_n$.

In step 5 the instance $A_j \cup B_j$ is implicitly interpreted as a $(k-1)$LIN instance over the $n-2$ variables $[n] \setminus \{1, j\}$. Assuming C has rate per sample λ (which in this setting means Poisson($\lambda n(n-1) \cdots (n-k+1)$) many examples), $A_j \cup B_j$ is now identically distributed to a planted (resp., random) $(k-1)$LIN instance with $n-2$ variables of rate per sample 2λ when s_1 equals (resp., does not equal) s_j.

A minor complication also arises in amplification. In Claim 8 the indices are now not independent as they are conditioned on being distinct leading to a slightly worse lower bound of $\eta(\alpha - (k-1)/n)^{k-1}$. Again this only affects asymptotically insignificant quantities in Theorem 11.

We do not know how to extend the proof of Theorem 2 to the model in which indices are chosen without repetition. The reason is that the search-to-distinguishing reduction does not preserve the balance of the secret. It remains balanced in the case $s_1 \neq s_j$ but loses its balance when s_1 equals s_j.

5.2 Extension to other finite fields

The kLIN problem naturally extends to other groups. When the group is a field $\mathbb{F}$, there is an alternative model in which the coefficients are not integers (representing the number of occurrences) but random nonzero field elements. In this $(k, \mathbb{F})$-LIN model the sample representing the left-hand side $a_1 x_{i_1} + \cdots + a_k x_{i_k}$ is specified by the sequence of coefficient-index pairs

$$\mathbf{i} = ((a_1, i_1), \dots, (a_k, i_k))$$

where $a_1, \dots, a_k$ are nonzero field elements and $i_1, \dots, i_k$ are coordinates in $[n]$, all sampled uniformly and independently (with replacement).

The *search* problem asks to find $\mathbf{s} \in \mathbb{F}^n$ from $(\mathbf{i}_1, \ell_1), \dots, (\mathbf{i}_m, \ell_m)$, whereas the *decision* problem asks to tell apart the planted distribution $(\mathbf{i}_1, \ell_1), \dots, (\mathbf{i}_m, \ell_m)$ from the null distribution $(\mathbf{i}_1, b_1), \dots, (\mathbf{i}_m, b_m)$ where b_i is a random element from $\mathbb{F}$.

Theorem 14. *There is an algorithm of complexity $O((m+n)|\mathbb{F}|^2)$ that, given access to a distinguisher for m $(k-1, \mathbb{F})$-LIN with advantage $1 - \delta/|\mathbb{F}|(n-1)$, solves $(k, \mathbb{F})$-LIN except with probability $\delta + \exp(-\Omega(m))$ from $4mn$ examples under any noise distribution.*

Proof. By Claims 3 and Claim 4, it is sufficient to prove the theorem in the Poisson sample model with rate per sample $\lambda = mn/(n(|\mathbb{F}|-1))^k$ for $(k, \mathbb{F})$-LIN search (which means Poisson($\lambda(n(|\mathbb{F}|-1)^k)$) many examples) and rate per sample $(|\mathbb{F}|/(|\mathbb{F}|-1)) \cdot \lambda$ for $(k-1, \mathbb{F})$-LIN distinguishing. The reduction is then implemented by

Algorithm S**:** On input a multiset C of examples and the value of $s_1 \in \mathbb{F}$,

Set $A = \{((a_2, i_2), \dots, (a_k, i_k); \ell - a_1 \cdot s_1) : ((a_1, 1), (a_2, i_2), \dots, (a_k, i_k); \ell) \in C\}$

Keep each example in A with probability $1/(|\mathbb{F}|-1)$ and discard otherwise.

3 For $j = 2$ up to n,
4 For v ranging over $\mathbb{F}$,
5 Set $B_j = \{((a_2, i_2), \dots, (a_k, i_k); \ell - a_1 \cdot v) : ((a_1, j), (a_2, i_2), \dots, (a_k, i_k); \ell) \in C\}$
6 If $D(A \cup B_j)$ accepts, set $\hat{s}_j = v$.
7 If all $\hat{s}_j$ have been uniquely set, output $\hat{\mathbf{s}}$.

The rate per sample of the reduction is as desired because the rates per sample of A and B_j are $\lambda/(|\mathbb{F}| - 1)$ and λ, respectively.

If $v = s_j$ then the examples in $A \cup B_j$ are identically distributed to planted $(k-1)$LIN because the label of every sample has been shifted by the discarded entry ($a_1 s_1$ for examples in A and $a_j v = a_j s_j$ for samples in B_j).

If $v \neq s_j$ then the examples in $A \cup B_j$ follow the null distribution. For examples in A the right-hand side follows the planted distribution $(a_2 s_{i_2} + \dots + a_k s_{i_k}) + e$. For examples in B_j this value is shifted by $a_1 \cdot (s_j - v)$ which is a uniform nonzero field element. Each example originates from A with probability

$$\frac{1/(|\mathbb{F}| - 1)}{1 + 1/(|\mathbb{F}| - 1)} = \frac{1}{|\mathbb{F}|}.$$

The shift is therefore zero with probability $1/|\mathbb{F}|$ and a nonzero random field element otherwise. Its distribution is uniform over $\mathbb{F}$ and therefore so is the label itself.

As the advantage is assumed to equal $1 - \delta/|\mathbb{F}|(n-1)$, D rejects at most a $\delta/|\mathbb{F}|(n-1)$ fraction of planted instances and accepts at most a $\delta/|\mathbb{F}|(n-1)$ fraction of null instances. By a union bound over all elements in $\mathbb{F}$ and $n-1$ locations, except with probability δ, line 6 triggers exactly when $v = s_j$ for all j. In that case $\hat{\mathbf{s}} = \mathbf{s}$.

Our algorithm works under the assumption that the value of the first secret s_1 is known. This assumption, however, can be removed by rerunning the algorithm with different values of s_1 over the field $\mathbb{F}$ (using the same set of samples) and taking the best answer. Therefore the algorithm time complexity is $O((m+n)|\mathbb{F}|^2)$ with $O(n|\mathbb{F}|^2)$ calls to the distinguisher. □

To decrease the probability of failure for the search algorithm, we can construct a distinguishing algorithm with a success rate close to 1. A distinguisher of success rate α can be amplified into one of the desired rate $1 - \delta/n|\mathbb{F}|$ at a multiplicative cost of $O((\log(n|\mathbb{F}|/\delta))/\alpha^2)$ in sample complexity. This is achieved through repetition and taking the majority of the predictions, followed by analysis with the Hoeffding bound.

6 Concluding remarks

6.1 On the sample complexity gap

In the constant advantage regime, our reduction solves kLIN from $O(mn)$ samples given a distinguisher for m samples for $(k-1)$LIN. The best known efficient search algorithm for kLIN requires $\Omega(n^{k/2})$ samples, while the best known distinguisher for $(k-1)$LIN requires $\Omega(n^{(k-1)/2})$ samples. If our reduction's sample complexity were improved by a $\sqrt{n}$ factor our reduction would match the optimal sample complexity for search to decision. Towards understanding this gap it is instructive to revisit one efficient search algorithm for kLIN.

Algorithm *Search***:** On input a multiset C of samples $(i_1, \dots, i_k; \ell)$,
1 Set $A_i = \{(i_2, \dots, i_k; \ell) \colon (i, i_2, \dots, i_k; \ell) \in C\}$ for every i.
2 For every pair $i \neq j$,
3 Set $M_{ij} = \ell_i \cdot \ell_j$ if the same sample $(i_2, \dots, i_k)$ appears in A_i and A_j with labels ℓ_i and ℓ_j.
4 Set $M_{ij} = 0$ if the samples in A_i and A_j are disjoint.
5 Derive $\hat{\mathbf{s}}$ from M by spectral analysis.

The implementation of step 5 is not important for our purposes. $\hat{\mathbf{s}}$ can be taken to equal the sign of the top eigenvector of M after rows with unusually many nonzero entries are removed (see e.g. [FO05, CRV15]).

Given $Kn^{k/2}$ samples for sufficiently large K, the output of this algorithm has $\Omega(1)$ overlap with the planted solution $\mathbf{s}$. The reason is that M can be modeled as a sparse "stochastic block model" matrix with respect to the partition induced by the $+1$ and -1 of $\mathbf{s}$.

Like our reduction S, algorithm *Search* leverages noisy information M_{ij} about $s_i s_j$ (equality of s_i and s_j) to recover an approximation of the secret $\mathbf{s}$. There is however an important difference.

The *Search* algorithm ensures robustness against noise by using all pairwise predictions $M_{ij} \approx s_i s_j$. In contrast, reduction S only uses information about the correlation of s_j with s_1. The reason the additional information is ignored by the reduction is that an adversarial distinguisher with success rate α can refuse to reveal any useful information for a $(1-\alpha)$ fraction of M's rows and columns. In contrast, the success of *Search* relies on the (almost) independence of the predictions M_{ij}. When α is on the order of $1/n$ as in the setting of algorithm *Search* an adversarial distinguisher yields essentially no information about $\mathbf{s}$.

6.2 The unique sample model

Our instance model allows repeated samples. We assumed that the kLIN noise (more generally, the internal randomness of P) is chosen independently among repeated samples. An alternative model is one in which samples cannot repeat. The two models are $O(mn^{-2})$ statistically close. Our results carry over assuming the advantage α is at least as large. The best low-degree kLIN distinguisher in the unique sample model, however, has advantage $\Theta(m^4 n^{-2k})$. Our reduction is sensitive to the choice of model in the low-advantage regime $m^4 n^{-2k} \ll \alpha \ll m^2 n^{-k}$.

6.3 Exact kLIN and refutation

We believe that our reduction is of interest even in the noiseless setting. Even though the search variant of noiseless kLIN can be solved by Gaussian elimination beyond the statistical threshold of $O(n)$ examples, the distinguishing algorithm is substantially simpler. It is an AC^0 algorithm, while Gaussian Elimination is believed to require more power than NC^1.

In the exact kLIN setting the distinguisher that seeks collisions is in fact a refuter. It rejects all planted kLIN instances but accepts an $\Omega(m^2 n^{-k})$ fraction of random ones. If the distinguisher in Theorem 1 is a refuter of advantage α, the sample complexity of the kLIN solver can be improved to $O(nm + n/\alpha^{(k-2)})$, economizing on an additional square factor in the advantage. Moreover the reduction can be implemented in TC^0 at a slight loss in parameters.

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